

Application Flow

Document 04

Status	Draft v1
Version	1.2 · 2026-09-24 22:26 · change log
Related	03 UI/UX , 05 Backend Schema

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The system has two loops that meet at the corpus:

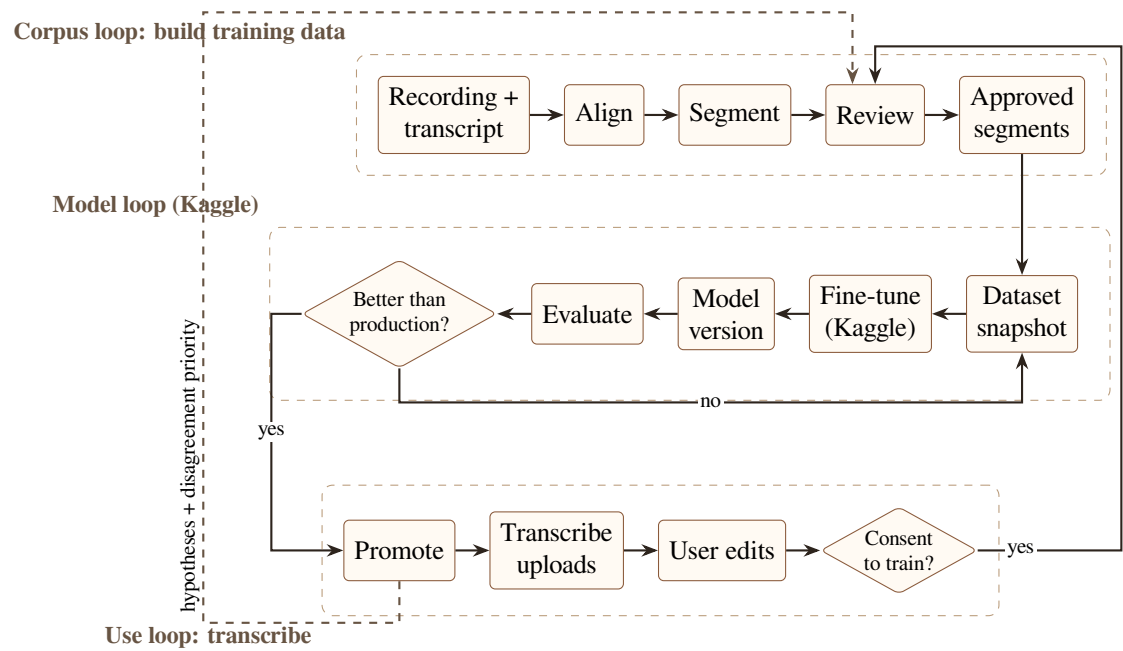


Figure 1: The corpus, model and use loops meet at the review queue.

1 Flow A: Import and align a transcribed recording

Changed in v1.2: see the change log.

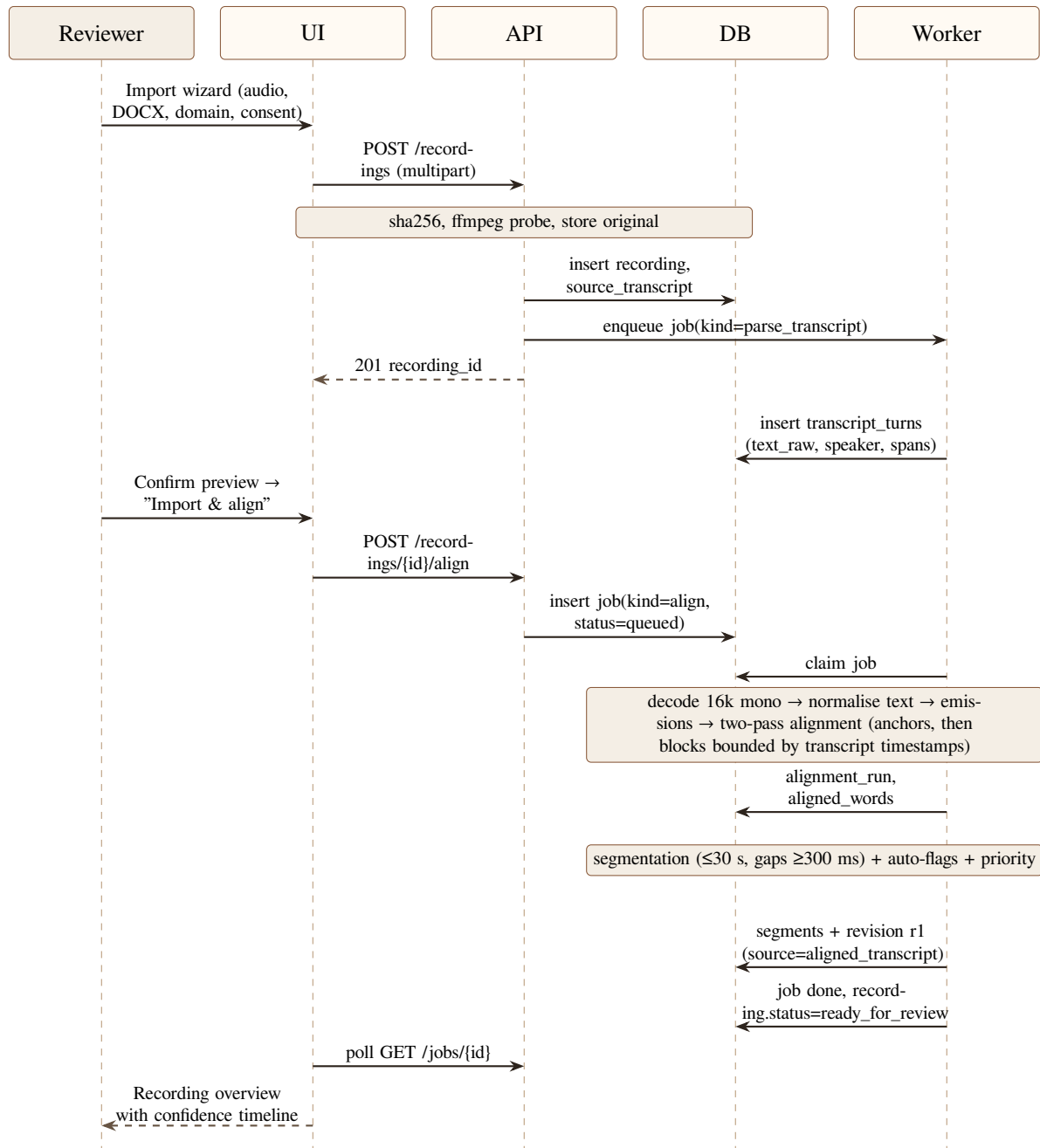


Figure 2: Flow A: import and align a transcribed recording.

Failure branches:

Condition	Result
Duplicate sha256	409 with a link to the existing recording
Transcript parses to 0 turns	Wizard step 3 shows an error; the user can import without a transcript (Flow B)
Alignment mean confidence < 0.3 for the whole recording	Job succeeds but the recording is flagged <code>alignment_suspect</code> (wrong transcript attached?); the overview shows a warning and offers “Re-align with other transcript” or “Fall back to VAD”

Condition	Result
Worker crash	Heartbeat expires and the job is re-queued (max 3 attempts, then failed)

2 Flow B: Import a recording without a transcript

POST /recordings without a transcript enqueues vad_segment: Silero VAD produces windows of ≤ 30 s, which become segments with an empty r1. If a production model exists, hypothesize jobs create hypotheses and r2 (source=model_hypothesis, clearly labelled). The recording then enters the same review queue. Priority comes from model confidence instead of alignment confidence.

3 Flow C: Review a segment

Each transition that changes content creates a new segment_revisions row. segments.current_revision_id moves forward, and old revisions are never deleted.

Sequence for “Approve & next”:

1. The UI sends POST /segments/{id}/revisions {base_revision_id, text, start_s, end_s, flags, spans, speaker_label, status:"approved"}.
2. The API checks base_revision_id == segments.current_revision_id. If it differs → 409.
3. In one transaction, the API inserts the revision, updates the segment (status, current_revision_id, bounds), and invalidates the cached audio slice if the bounds changed.
4. The API returns the next segment from GET /review/queue: ordered by priority DESC, ordinal ASC and skipping approved|rejected.
5. The UI loads it and autoplays.

“Run model” (R): POST /segments/{id}/hypothesis runs inline if the model is loaded (< 3 s on CPU for a ≤ 30 s clip with whisper-small int8), otherwise it enqueues a job. The result is stored in hypotheses. The editor text is **unchanged**. The diff and WER are displayed, and segments.priority is recomputed with model disagreement.

4 Flow D: Build a dataset snapshot and train on Kaggle

Rules:

- The test split is attached from the frozen test_v1 definition and is never re-sampled.
- The worker refuses to register a model whose run_manifest.dataset_sha256 is unknown. This prevents mystery models.
- The notebook is thin: !pip install -e /kaggle/input/transcribe-src followed by python -m transcribe.finetune.whisper_train --config ..., so the training code is version-controlled in this repo.

5 Flow E: Evaluate and promote

1. POST /evaluations {model_id, dataset_id, split} enqueues a job.
2. The worker transcribes each segment with the model (CT2 int8, the same path as production), normalises, computes metrics, and writes eval_runs + eval_results.
3. The report screen shows metrics and the worst segments. “Send to review” pushes those segments to the top of the review queue (priority = 1.0). Worst segments are often label errors, not model errors.

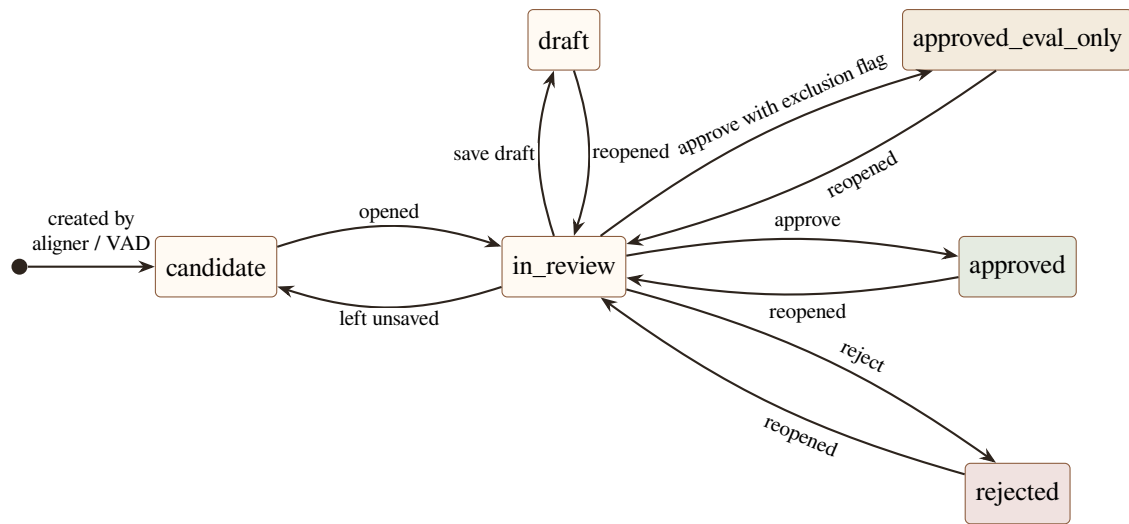


Figure 3: Flow C: segment review state machine. Every content change creates a new revision.

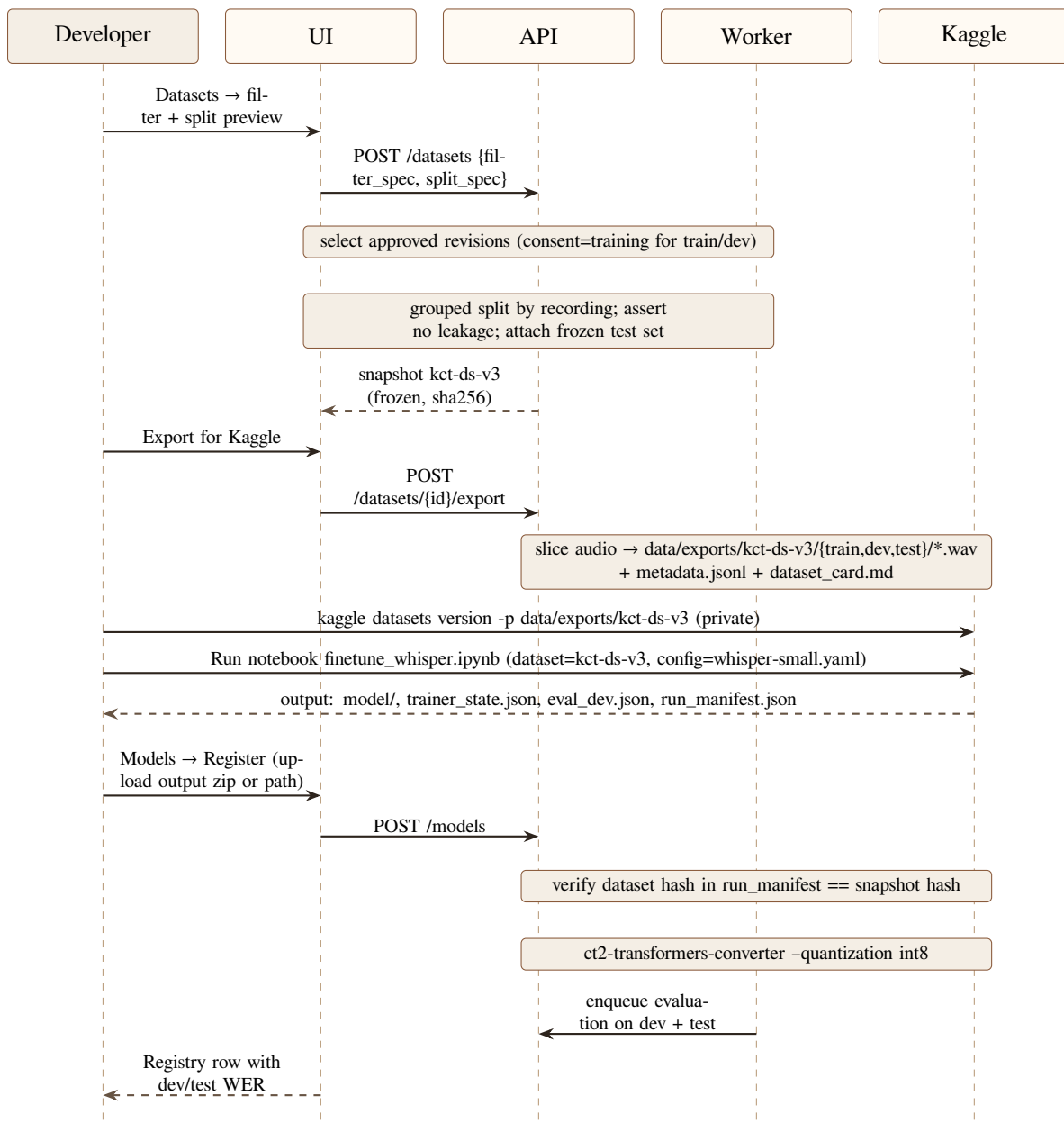


Figure 4: Flow D: build a dataset snapshot and train on Kaggle.

- Promotion gate (enforced by the API; override needs `force=true` + a note): the candidate's **test WER $\leq$ production test WER**, **hallucination rate is not worse by more than 10%**, and **RTF ≤ 0.5** .
- `POST /models/{id}/promote` sets `status=production`, retires the previous one, and the Transcribe service reloads the model.

6 Flow F: Transcribe a file

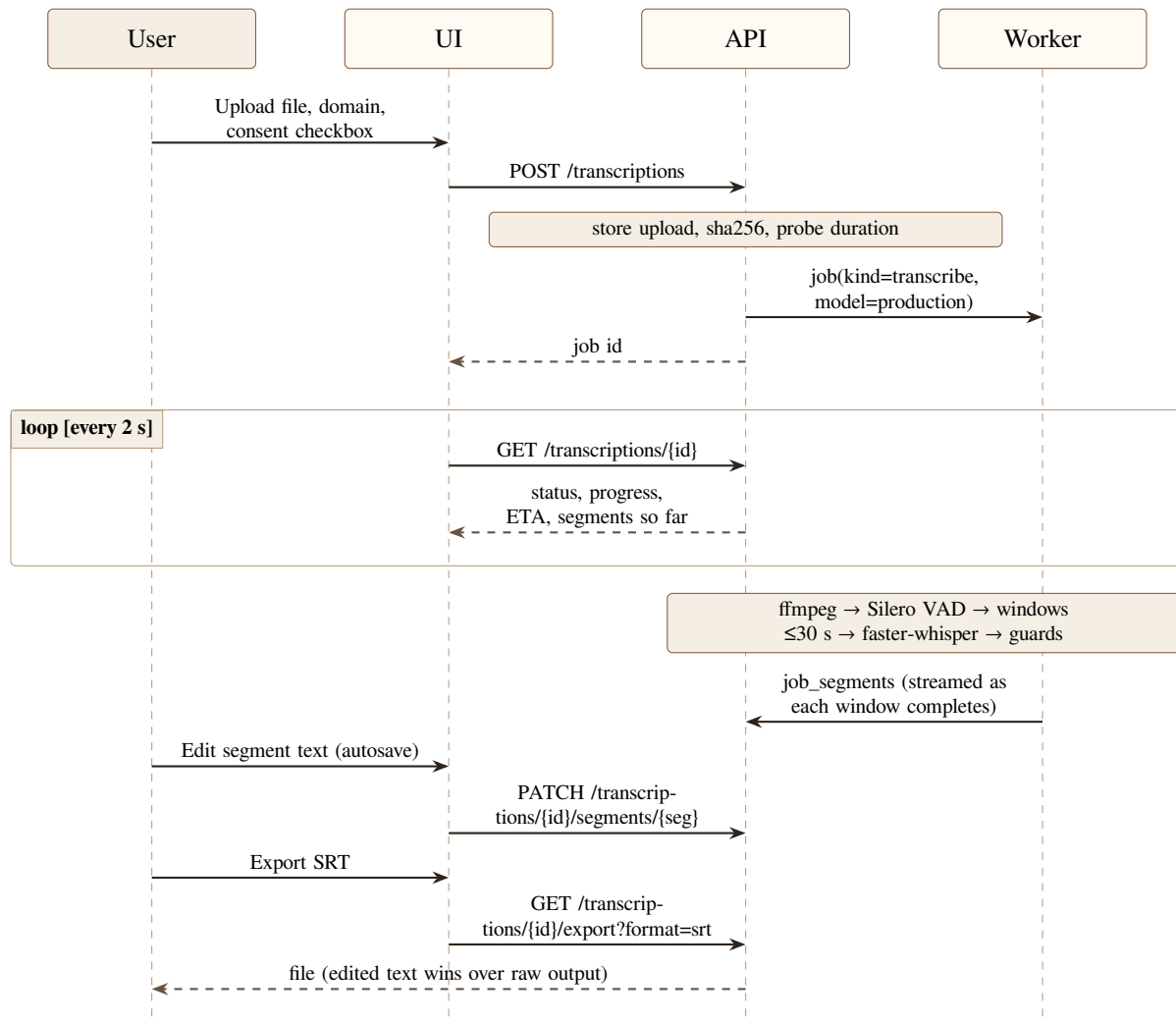


Figure 5: Flow F: transcribe a file.

Optional “Promote edits to corpus” (only if consent was given at upload):

- Each edited job segment becomes a corpus segment (`origin=inference_correction`) with `r1` = the model output (`source=model_hypothesis`) and `r2` = the user's edit (`source=human`, `status=candidate`).
- These enter the review queue at high priority. They are the most valuable training examples, because they are exactly where the model failed.
- Unedited segments are **not** promoted: unverified model output must not become training data.

7 Flow G: Active learning cycle (v2)

- Once a week, run the production model over all un-reviewed segments (recordings without transcripts, promoted jobs).
- Priority = model uncertainty (mean token log-prob, low-confidence word count) plus disagreement with

the aligned text where it exists.

3. Review the top-N segments.
4. Build a snapshot, fine-tune, evaluate, and apply the promotion gate.
5. Track learning curves: test WER vs. hours of approved data per snapshot. This data answers the question “how much more data do we need?”.

8 Error-handling summary

Where	Error	Behaviour
Upload	Unsupported codec / corrupt	422 with the ffmpeg message; nothing is stored
Any job	Exception	status=failed, error stored, retry button; attempts ≤ 3
Review save	Stale base revision	409, conflict banner, nothing lost
Model load	CT2 files missing	The model is marked broken; Transcribe falls back to the previous production model with a banner
Kaggle registration	Dataset hash mismatch	Registration rejected with the expected vs. found hash
Disk full	Write failure	The job fails cleanly; the UI shows the storage usage panel

Change log

Times are East Africa Time (EAT, UTC+3). Each entry names the affected section, quotes what it said before and summarises what it says now, and links to the previous text: on GitHub at the exact commit and lines, and as an archived PDF of the previous version.

Version	Date (EAT)	Commit	Summary
1.0	2026-09-24 00:36	053e09d	First LaTeX version (converted from the Markdown draft of 2026-09-23).
1.2	2026-09-24 22:26	this commit	Flow A diagram: alignment step renamed to match TRD §4.3 v1.2.

v1.2 • Flow A: Import and align (Figure 2)

Previous text: [v1.1 source, figures/seq_import.tex line 33](#) · [v1.1 PDF](#)

Before. Worker step: “decode 16k mono → normalise text → emissions → turn-anchored alignment”.

Now. “... → emissions → two-pass alignment (anchors, then blocks bounded by transcript timestamps)”, as validated in the pilot ([TRD §4.3](#)).